1 Assignment 3 - Logistic regression and Basis Function Expansion

Visualization of the data shows that there are values equal to zero in the measures, including plasma glucose which is one of the predictors of interest in this assignment. `Glucose = 0` is not a feasible value, but rather representing a missing value. All cases (n=5) with plasma glucose equal to zero were excluded from the data before the regression analysis.

1.1 Scatterplot diabetes vs age

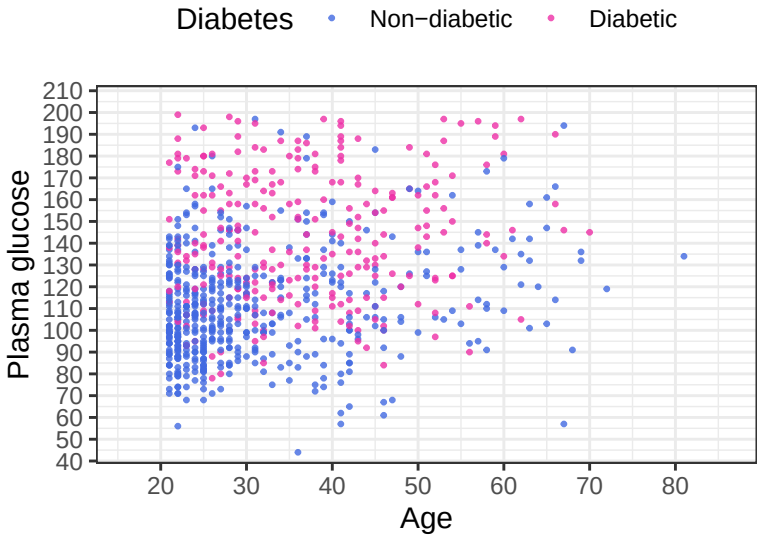


Figure 1: Plasma glucose levels in relation to age. Blue dots represent non-diabetic subjects, the pink dots diabetic subjects.

Plotting plasma glucose levels vs. age shows that the two response groups, *Non-diabetic* and *Diabetic*, show a substantial overlap. `Diabetes` would therefore not be easily predicted by *Age* and *Glucose* in a standard logistic regression model which would not be able to derive a straight line clearly separating the groups.

1.2 Logistic regression model

The vector form of the probailistic equation for a logistic regression model is

$$P(Y = 1 \mid x) = \text{sigmoid}(\%theta^T x) \tag{1}$$

where the sigmoid function is

$$sigmoid(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}} \tag{2}$$

For *Diabetes* as the binary target/response (y), and *Glucose* and *Age* as features/predictors (x1, x2):

$$P(Y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 \cdot x_1 + \theta_2 \cdot x_2)}} \tag{3}$$

Training a logistic regression model on the (entire) provided data set, using glm() in R, results in the coefficients $\theta_0 = -6.2252$, $\theta_1 = 0.0384$, and $\theta_2 = 0.0231$, all with significant p values, and the corresponding probabilistic equation is then

$$P(Y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-6.2252 + 0.0384x_1 + 0.0231x_2)}} \tag{4}$$

The training misclassification error for this model (with r = 0.5) is 0.26, i.e approximately one out of four observations are misclassified.

1.3 Scatterplot colored by predicions

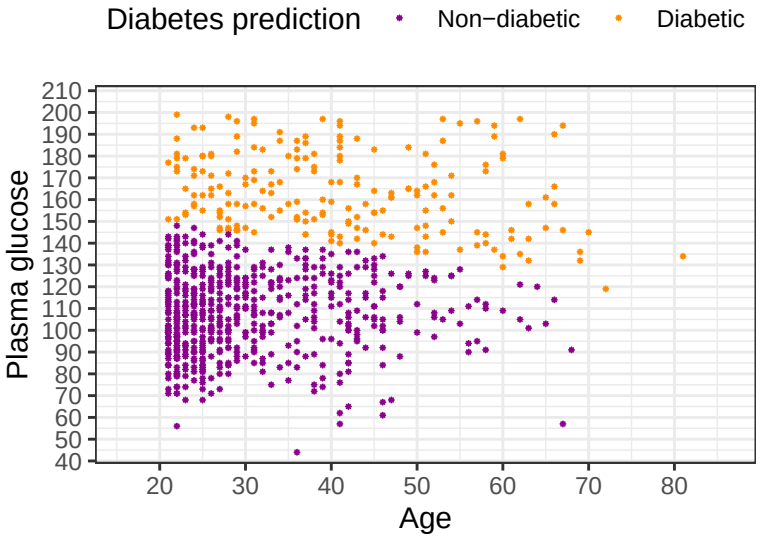


Figure 2: Plasma glucose levels in relation to age. Purple stars represent subjects predicted to be non-diabetic, the orange stars subjects predicted to be diabetic.

Comparing the predicted diabetes classifications to the actual diabetes status as shown in Figure 1 it looks like there will be some misclassifications, possibly more so for subjects in the lower, middle part of the scatter, where diabetic an non-diabetic subjects are overlapping, and for the non-diabetic elderly subjects.

1.4 Decision boundary equation

Setting the probabilistic equation in Equation 4 equal to `r` (here `= 0.5`) results in this approximate decision boundary equation, with the fitted coefficients and $x1 = Glucose$ and $x2 = Age$:

$$-6.2252 + 0.0384 \cdot Glucose + 0.0231 \cdot Age = 0 \tag{5}$$

Solving this for *Glucose* gives the approximate linear decision boundary line

$$Glucose = 161.97 - 0.6 \cdot Age \tag{6}$$

1.5 Scatterplot with decision boundary line for r = 0.5

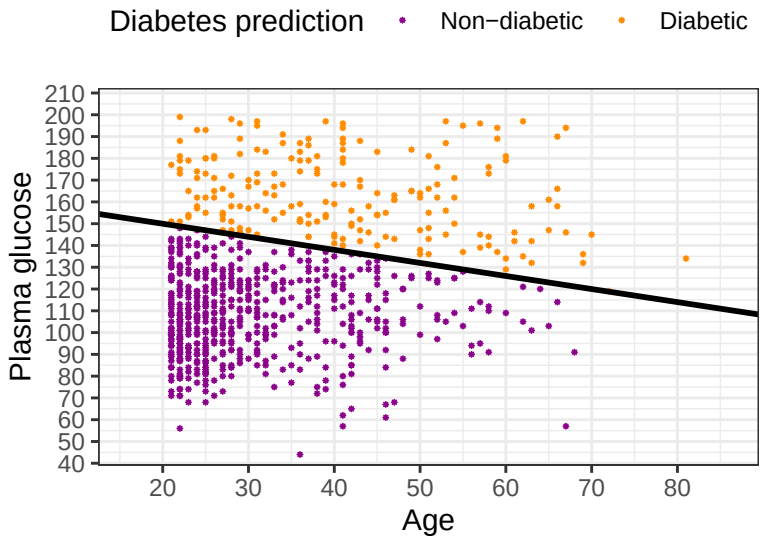


Figure 3: Predicted diabetes classifications with threshold r = 0.5. The added black line is the decision boundary line.

The boundary line means that there will likely be misclassifications both ways in the mixed "middle" group. The elderly non-diabetics will risk being misclassified as diabetic according to the negative slope of the linear boundary imposing increasingly harder restrictions on acceptable plasma glucose level for a non-diabetic subject.

1.6 Scatterplot with decision boundary line for r = 0.2 and r = 0.8

The equation for the linear decision boundary line for two predictors/features can generically be expressed as

$$x_1 = \frac{\log\left(\frac{r}{1-r}\right) - \theta_0}{\theta_1} - \frac{\theta_2}{\theta_1} x_2 \tag{7}$$

which means the intercept changes with r, the slope does not.

With r = 0.2 are more individuals predicted to be diabetic, with r increased to 0.8 instead fewer individuals. The training misclassification errors are 0.36 and 0.31 resepectively for the 0.2 and 0.8 cases, both higher than for the 0.5 cutoff case.

1.7 Basis function expansion

Fitting a new logistic regression model after adding new features

$$z_1 = x_1^4, \quad z_2 = x_1^3 x_2, \quad z_3 = x_1^2 x_2^2, \quad z_4 = x_1 x_2^3, \quad z_5 = x_2^4$$

results in a much more complex model but a training misclassification error (with r = 0.5) of 0.24 which is slightly better than the linear model. The decision boundary is now not linear but curvilinear, changing the appearance of the Glucose vs Age scatterplot (see Figure 5). The most obvious difference from the previous model is that older subjects are "allowed" to have higher plasma

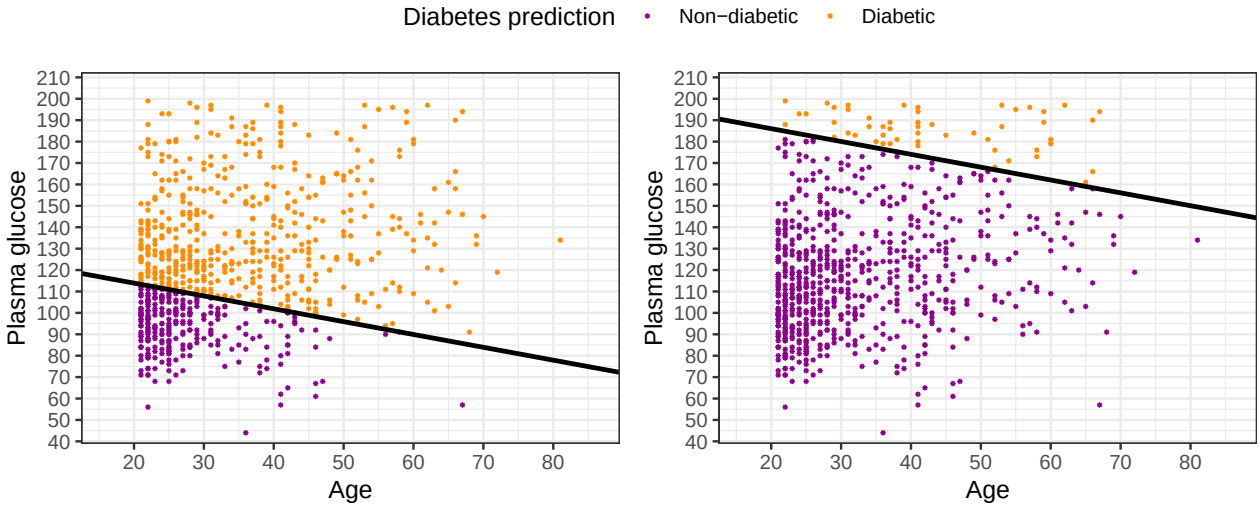


Figure 4: Predicted diabetes classifications for thresholds r = 0.2 (left) and r = 0.8 (right). Black lines indicate the decision boundaries.

glucose levels without being predicted to be diabetic, which is a pattern that can be discerned also in the actual data.

The *theta3* to *theta7* coefficients (for the *z1* to *z5* terms) are very small (magnitude $10^{-7} or less), and individually non-significant (p values above 0.6), but they are multiplied with numerically high values for *Glucose* (raised to up to 4) in the polynomial, and the AIC reduction suggests an improved overall fit for the model.

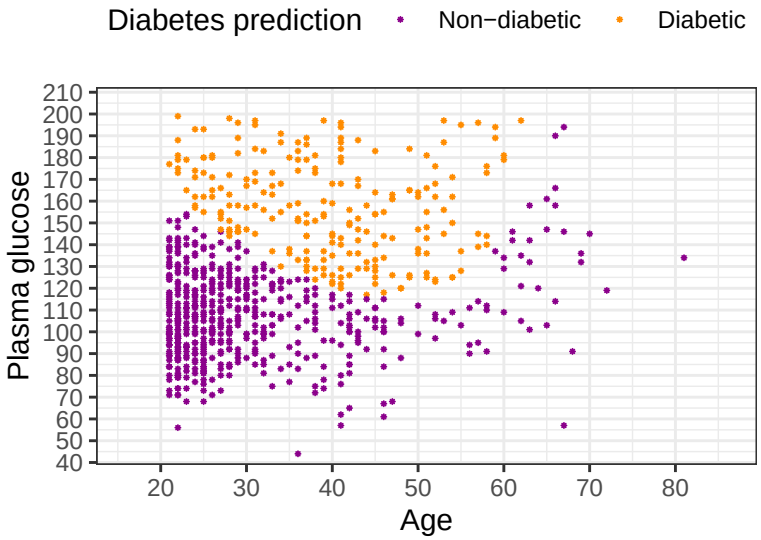


Figure 5: Predicted diabetes classifications for the extended logistic regression model with polynomial basis function expansion.